

GARLIC - Genomic Autozygosity Regions
Likelihood-based Inference and Classification v1.1.6
User Manual

Zachary A Szpiech

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1 Introduction

Extended tracts of homozygosity in individual genomes manifest as a result of haplotypes inherited identical by descent (IBD) from both biological parents. These runs of homozygosity (ROH) are important genomic features, with their length distributions and genomic locations informative about population history and useful for the mapping of recessive loci contributing to Mendelian and complex diseases. However, precisely because the length distribution of ROH is affected by population history, a simple one-size-fits-all genotype-counting approach to inferring ROH in multiple populations is ill-advised. Here, we present a model-based method and accompanying software package for inferring ROH in microarray-derived SNP genotype that incorporates population-specific parameters and genotyping error rates as described in Pemberton *et al.* (2012). Version 1.1.0 includes several updates, including a method for reweighting LOD scores by gaps between SNPs, the option to report ROH lengths in cM, and the ability to provide per-genotype error rates.

If you use GARLIC please cite the following articles,
ZA Szpiech, A Blant, TJ Pemberton. (2017) GARLIC: Genomic Autozygosity Regions Likelihood-based Inference and Classification. Bioinformatics doi: 10.1093/bioinformatics/btx102.

TJ Pemberton, et al. (2012) Genomic patterns of homozygosity in worldwide human populations. American Journal of Human Genetics, 91, 275 – 292.

2 Obtaining GARLIC

NOTE: In order to successfully run, windows users must have `ann_figtree_version.dll` and `figtree.dll` (provided) in the same folder as `garlic.exe`.

GARLIC pre-built binaries and source code are available at <https://github.com/szpiech/garlic>. To compile from source, change directories to the `src/` directory and type `make`; the Makefile detects the platform from `uname`, so no editing is needed. GARLIC depends on the zlib library (<http://www.zlib.net/>) and on POSIX threads; on Linux and macOS both are provided by the system and the Makefile simply links `-lz` and `-pthread`. GNU GSL and FIGTree were required by earlier versions and are no longer used: FIGTree was replaced by a direct Gaussian sum, and GSL by `src/garlic-math.cpp`, which reproduces the routines GARLIC used bit for bit. A win32 implementation of zlib is included, and is also available at <http://gnuwin32.sourceforge.net/packages/zlib.htm>. The windows version of GARLIC was built using a MinGW environment (<http://www.mingw.org/>).

3 Basic Usage

GARLIC can be run by executing the following command.

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --auto-winsize  
--out example
```

The example data is derived from human genotype chip data with hg18 coordinates (`--build hg18`), and a priori we do not know how large to make our window so we use the built-in window size selection algorithm (`--auto-winsize`). All output files will be named starting with 'example' (`--out example`). This produces the following files:

```
example.error  
example.log
```

```
example.60SNPs.kde
example.freq.gz
example.roh.bed
```

If the automatic window size selection algorithm fails, you can output the KDEs of the LOD score distribution for multiple window sizes (without calling ROH) by using the `--winsize-multi` argument, i.e.

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize-multi
30 40 50 60 70 80 90 --out example
```

This will generate KDEs for your inspection. Once you've chosen a window size, you should rerun garlic specifying that size. For example, if you choose a window size of 60 SNPs, then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example
```

If you already know what LOD score cutoff to use (say you are analyzing more individuals from a previously studied population), you can use the `--lod-cutoff` argument, i.e. if your LOD score cutoff is known to be 2.5 then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example --lod-cutoff 2.5
```

If you already know what size thresholds to use for size classification (say you are analyzing more individuals from a previously studied population), you can use the `--size-bounds` argument, i.e. if your size thresholds are known to be 500000 and 1000000 for the boundaries between short/med and med/long, respectively, then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example --lod-cutoff 2.5 --size-bounds 500000 1000000
```

To run the wLOD, you must provide a map file and give the `-weighted` flag:

```
garlic --tped example.tped.gz --tfam example.tfam --map example.map.gz --weighted --build
hg18 --winsize 60 --out example --error 0.001
```

Other command line arguments are described below.

4 Statistics implemented

We previously advocated the application of a logarithm of the odds (LOD) score measure of autozygosity applied in a sliding-window framework to infer ROH in high-density SNP genotype data (Pemberton *et al.*, 2012). Fundamentally, for SNP k in individual i , this method calculates the log-likelihood ratio of observed genotype $G_{i,k}$ under the hypotheses of autozygosity and non-autozygosity, incorporating an assumed genotype error rate ϵ and population-specific allele frequencies. The LOD score of window w in

individual i is then calculated as the sum of the log-likelihood ratios of the K SNPs in the window:

$$LOD(w, i) = \sum_{k=1}^K \log_{10} \left(\frac{Pr[G_{i,k}|X_k = 1]}{Pr[G_{i,k}|X_k = 0]} \right). \quad (1)$$

Here, $Pr[G_{i,k}|X_k = 1]$ is the probability of observing genotype $G_{i,k}$ under the hypothesis of autozygosity ($X_k = 1$), and $Pr[G_{i,k}|X_k = 0]$ is the probability of observing genotype $G_{i,k}$ under the hypothesis of non-autozygosity ($X_k = 0$). For a biallelic locus with alleles A and B that have population frequencies p_A and p_B and a genotype error rate ϵ , the genotype probabilities under the autozygosity and non-autozygosity hypotheses are given in Table 1.

Calculating $LOD(w, i)$ for all windows in all individuals in a given sample set, examination of the distribution of scores shows clear bimodality, with windows in the left-hand mode supporting the hypothesis of non-autozygosity and those in the right-hand mode supporting the hypothesis of autozygosity. The area under the autozygous mode decreases with increasing window size until it disappears, potentially reflecting the window size beyond which window length is often longer than ROH length leading to the inclusion of non-autozygous regions in the $LOD(w, i)$ calculation that mask the presence of autozygosity. A logical window size to use for ROH detection is therefore the largest window size where the distribution of $LOD(w, i)$ is bimodal, with windows defined as autozygous if their $LOD(w, i)$ is greater than the local minimum between the two modes. Contiguous autozygous windows are subsequently joined to define ROH.

If the weighted flag is used, wLOD scores will be calculated as

$$wLOD(w, i) = \sum_{k=1}^K \log_{10} \left(\frac{Pr[G_{i,k}|X_k = 1]}{Pr[G_{i,k}|X_k = 0]} \right) \times Corr(p_k, [p_1, p_K]) \quad (2)$$

$$\times Pr[norecombination|[g_{k-1}, g_k]] \quad (3)$$

$$\times Pr[nomutation|\mu, [p_{k-1}, p_k]], \quad (4)$$

where

$$Corr(p_k, [p_1, p_K]) = \frac{1}{\sum_1^K LD_{k,i}}, \quad (5)$$

$$Pr[norecombination|[g_{k-1}, g_k]] = e^{-2M(g_k - g_{k-1})}, \quad (6)$$

and

$$Pr[nomutation|\mu, [p_{k-1}, p_k]] = e^{-2M\mu(p_k - p_{k-1})}. \quad (7)$$

$LD_{k,i}$ is computed as r^2 if data are phased and as HR^2 if data are unphased. g_i and p_i are the genetic distance and physical distance at marker i , respectively.

Table 1: Probability model for genotypes under autozygosity and non-autozygosity.

Observed genotype G_k	$Pr[G_{j,k} X_k = 1]$	$Pr[G_{j,k} X_k = 0]$
AA	$(1 - \epsilon)p_A + \epsilon p_A^2$	p_A^2
AB	$2\epsilon p_A p_B$	$2p_A p_B$
BB	$(1 - \epsilon)p_B + \epsilon p_B^2$	p_B^2
Missing	1	1

5 Program Options

Using the command line flag `--help`, will print a help dialog with a summary of each command line option. The full reference is reproduced in Appendix A, which is generated from those same help strings by `make docs` so that it cannot drift from the program.

5.1 Input Files

Genotypes are read from either a VCF (`--vcf`) or a TPED/TFAM pair (`--tped/--tfam`, see <http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml#tr>). The two are mutually exclusive. Either may be read gzipped (<http://www.gzip.org/>) without decompressing first, and a VCF may also be BGZF, as written by `bgzip` or `bcftools`. Consecutive loci are assumed to be in order with respect to their physical position, and all records for a chromosome must be contiguous — what `bcftools sort` produces. `garlic` computes allele frequencies over every sample in the file, so different populations should be analysed separately; `--pop` supplies the population label that appears in the output.

5.1.1 VCF input

Biallelic SNVs carrying a `GT` field are used. Multiallelic sites, indels, symbolic ALT alleles (`*`, `<NON_REF>`) and records with no `GT` in `FORMAT` are skipped, and the counts are written to the log rather than passed over silently. The `FILTER` column is reported either way and only acted on under `--vcf-pass-only`, since it records the variant caller’s thresholds rather than `garlic`’s.

A VCF states things a TPED cannot, and `garlic` checks them: ploidy is per call, so anything other than a diploid genotype is an error naming the site and the sample; phase is per call, so `--phased` requires `|` everywhere rather than silently mis-reading a `/`; and samples are named in the `#CHROM` line, so `--tfam` is not accepted alongside `--vcf` — it could only disagree. Use `--pop` for population labels; omitting it labels every sample `unknown` and warns, because the allele frequencies really are pooled over all of them.

The counted allele is ALT. For the same genotypes a TPED counts whichever allele it saw first, so the frequency column may come out as $1 - f$ relative to a TPED run. This changes no ROH call, and a `.freq.gz` file remains interchangeable between the two readers: `readFreqData` compares the `ALLELE` column against the reader’s own coding and flips the frequency when they differ.

Genetic map positions are not part of a VCF, so `--weighted` and `--cm` still require `--map`.

Per-genotype genotype qualities are read from the `FORMAT` column with `--gl-type`. This is the only correct way to use a VCF’s `PL` or `GL`: a VCF normalises both so the called genotype’s value is exactly 0, so that single value is 0 for every call, confident or not, and the information is in how much worse the alternatives are. `garlic` therefore reads the whole array and computes

$$P(g) \propto 10^{-\text{PL}_g/10}, \quad \text{error} = 1 - \frac{P(\text{called})}{\sum_g P(g)},$$

with `GL` entering the same expression as `PL` = -10 GL . This reproduces the `GQ` that encodes the same confidence: a `PL` of (0, 30, 60) gives 0.001, exactly `GQ` 30. If the requested field is absent the run stops and the message names whichever of `GQ`, `PL` and `GL` the record does carry.

The companion script `vcf2tped.pl` is deprecated. It writes a TPED plus a `tgls` file, and a `tgls` file holds one value per genotype, which cannot represent a normalised `PL` or `GL`; `garlic` now rejects such a file rather than computing with it.

5.1.2 File formats

tped/tfam (required):

<http://zzz.bwh.harvard.edu/plink/data.shtml#tr>

map (required when `-weighted` or `-cm` are set):

<chr> <snpid> <genetic pos> <physical pos>
<http://zzz.bwh.harvard.edu/plink/data.shtml#map>
(negative positions will cause errors)

tgls (optional; legacy — see the note on `vcf2tped.pl` above):

<chr> <snpid> <unused> <physical pos> <GL ind1> <GL ind2> ... <GL indN>

One value per genotype. Under `--gl-type PL` or `GL` these are read as the probability that the genotype is *correct*, which is *not* what a VCF's PL and GL fields hold; a file of VCF-derived values is rejected with an error. `--gl-type GQ` is the one form whose VCF values can be used here directly.

pop (optional; with `--vcf`, the only way to set population labels):

<sample id> <population> [sex]

Matched by sample ID, never by position. Note the column order is the opposite of a TFAM's first two columns; a TFAM-ordered file is diagnosed as such. The optional third column uses PLINK sex coding and only makes the sex-chromosome warning specific. freq (optional, output during computation, can be generated without running whole pipeling with `-freq-only`):

CHR SNP POS ALLELE FREQ
<chr> <locus ID> <allele> <freq>

centromere file (required if no `-build` is set):

<chr> <centromere start> <centromere end>

5.1.3 --tped

Use `--tped <string>` to specify a .tped (transposed PLINK; Purcell *et al.* (2007)) file (see <http://zzz.bwh.harvard.edu/plink/data.shtml#tr> for exact specifications) containing genetic variant information.

5.1.4 --tped-missing

Use `--tped-missing <char>` to specify the missing data code in the .tped file. Default is 0.

5.1.5 --tfam

Use `--tfam <string>` to specify a .tfam (transposed PLINK; Purcell *et al.* (2007)) file (see <http://zzz.bwh.harvard.edu/plink/data.shtml#tr> for exact specifications) containing sample information.

5.1.6 --tgls

Use `--tgls <string>` A tgls file containing per-genotype likelihoods. Formatted: `jchrj jsnpidj junusedj jphysical posj jGL ind1j jGL ind2j ... jGL indNj`. Default: none.

Per-genotype likelihoods should be given as either a phred-scaled (`--gl-type PL`) or a $\log_{10}$ -scaled (`--gl-type GL`) representing the probability that the genotype is correct or a phred-scaled likelihood that the genotype is wrong (`--gl-type GQ`). For example, if $p = 0.999$ is the probability that the genotype is correct, then $PL = -10 \log_{10}(p) = 0.00434511774018$, $GL = \log_{10}(p) = -0.000434511774018$, and $GQ = -10 \log_{10}(1 - p) = 30$.

5.1.7 --centromere

Use `--centromere <string>` to specify a file containing custom centromere boundaries formatted as

```
<chr> <start> <end>
```

5.1.8 --freq-file

`--freq-file <string>` A file specifying allele frequencies for all variants. File format:

```
CHR      SNP      POS      ALLELE  FREQ
<chr>    <locus ID>    <pos>    <allele>    <freq>
```

By default, this is calculated automatically from the provided data.

5.1.9 --map

`--map <string>` Provide a scaffold genetic map, sites that aren't present within this file are interpolated. Sites outside the bounds are filtered. This is required for wLOD calculations and any runs for which you wish to report ROH in units of cM. Formatted: `|chr| |snpid| |genetic pos| |physical pos|`. <http://zzz.bwh.harvard.edu/plink/data.shtml#map> (negative positions will cause errors)

5.2 Output Files

`garlic` produces several files as output. The `.log` file will record the runtime parameters and `.error` will record any errors that occur; `.error` is created only if something is actually written to it. A `.params.json` file records the effective value of every flag together with the values resolved during the run (selected window size, LOD cutoff, size class boundaries and random seed), and can be replayed with `--load-params`.

A gzipped `.freq` file will be output giving the allele frequencies for each locus in the dataset. Formatted as

```
CHR      SNP      POS      ALLELE  FREQ
<chr>    <locus ID>    <pos>    <allele>    <freq>
```

Kernel density estimation of the LOD score distribution is output to `.<winsize>SNPs.kde` formatted as

```
<LOD score> <density>
```

ROH calls are output to `.roh.bed`, which are formatted as UCSC BED files. There is one track per individual (all individuals in the same file) and the data are formatted as

```
<chr> <ROH start> <ROH end> <size class> <ROH length> <placeholder> <
  placeholder> <placeholder> <RGB track color>
```

Following the BED specification, `<ROH start>` is 0-based and `<ROH end>` is exclusive, so `<ROH end>` minus `<ROH start>` equals `<ROH length>`. Note that under `--cm` the boundaries remain physical positions while `<ROH length>` is reported in cM, so that identity does not hold in that mode. Versions before this one wrote `<ROH start>` as a 1-based position, making every tract appear one base short in a genome browser; coordinates from those versions are shifted by one relative to these.

When requested, raw LOD scores are also output. This file contains one row per individual (corresponding to the order in the `.tfam` file), and LOD scores for each window are given on the columns.

<LOD score window 1> <LOD score window 2> ... <LOD score window N>

5.2.1 --out

Use `--out <string>` to provide a base name for an output file. This will be used in place of <outfile> above. Default value is outfile.

5.3 Controlling GARLIC

The remaining flags control window construction, LOD score calculation, cutoff selection, size classification, reproducibility and output. They are listed with their types and defaults in Appendix A, which `make docs` generates from `garlic`'s own help strings. This section previously restated them by hand, which is how it came to document `--auto-winsize-size` – a flag the program has never accepted – and to miss every flag added since.

6 Change Log

24AUG2017 - v1.1.6 Modified behavior of `--cm` flag based on user feedback.

When this flag is set ROH lengths and length classifications will be reported in cM but ROH boundaries will be reported in bps.

18JUL2017 - v1.1.5 Introduced `--ld-subsample` argument. If set > 0 , will use only a subset of individuals for computing LD statistics when calculating wLOD scores.

18MAY2017 - v1.1.4 If a freq file is provided, GARLIC now ensures that internal coding of alleles is consistent with the allele specified in the file. Improved handling of loci with 100% missing data. Improved reading of gzipped freq files. Improved lod score cutoff discovery. Removed default setting for `--winsize`, which must now be specified. Now allows `--auto-winsize` to be used with `--winsize-multi`, will choose best from the list.

12MAY2017 - v1.1.3 adds a new option for `--gl-type` called GQ, which is a phred-scaled probability that the genotype is wrong. If $p = 0.999$ is the probability that the genotype is correct, then $GQ = -10 \cdot \log_{10}(1-p) = 30$. The default for `--gl-type` is now set to none, so if you provide a TGLS file you will be required to choose between GQ, PL, and GL.

08MAY2017 - v1.1.2b fixes a bug that causes a crash when `--overlap-frac` is set to 0. Setting to zero now chooses the smallest sensible overlap fraction ($1/\text{winsize}$). If you wish to let `garlic` automatically guess overlap fraction, use the new command line option: `--auto-overlap-frac`. Also includes an iteration counter to monitor progress of GMM portion of the code.

07MAY2017 - v1.1.2 fixes a bug that causes a crash when a tglS file indicates a genotype has 0 likelihood. Add progress bars for LD calculations and LOD score calculations. Improve efficiency of tglS file loading.

06MAY2017 - v1.1.1a fixes a bug introduced with 1.1.1 that caused crashed when --resample is used.

05MAY2017 - Update to version 1.1.1. For some large datasets (e.g. WGS), KDE can take extremely long to complete, even when only a subset of individuals are considered. Here we introduce some changes to speed this calculation up. Instead of passing all LOD score windows to the KDE function, we now pass only non-overlapping windows. This reduces the number of points sent to the function by a factor of WINSIZE. If you wish to retain all LOD scores, you may set the --no-kde-thinning flag. We also change the default number of random individuals for KDE (--kde-subsample) to 20, which still may be set to 0 to use all individuals.

We also introduce a lower bound on the size of ROH reported. Now, an ROH will not be called unless it contains at least $WINSIZE * OVERLAP_FRAC$ number of SNPs. For example, for a WINSIZE (--winsize) of 100 and an OVERLAP_FRAC (--overlap-frac) of 0.25 the smallest possible ROH that will be reported will be 25 SNPs long.

Finally, if --overlap-frac is set to 0, GARLIC will attempt to guess a good choice.

02MAY2017 - Update to version 1.1.0a. Bug fixed that caused crashes when using TGLS file.

28APR2017 - Update to version 1.1.0. Includes several updates, including a method for reweighting LOD scores by gaps between SNPs, the option to report ROH lengths in cM, and the ability to provide per-genotype error rates.

25OCT2016 - Update to version 1.0.1. This update fixes a bug where ROH that extended upto the end of the chromosome failed to get assembled and reported. This update also introduces a new command-line flag --overlap-frac which is designed to reduce false positive calls by requiring a SNP be covered by at least OVERLAP_FRAC (default 0.25) proportion of high scoring windows to be included in an ROH call. This helps to reduce false positive calls near the boundaries of true ROH.

05MAY2016 - Initial release of GARLIC v1.0.0

A Command line reference

Generated from `garlic`'s own help strings by `make docs` in `src/`; do not edit `flags.tex` by hand.

`--M <int>` The expected number of meioses since a recent common ancestor for `--weighted` calculation.
Default: 7

`--auto-overlap-coef <double1> ... <doubleN> <slope> <intercept>`: coefficients of $\text{overlap\%} = \text{slope} * \log(\text{density}) + \text{intercept}$, used by `--auto-overlap-frac`. The defaults (6.375 63.888) are an empirical fit to human SNP-array data.
Default: 6.375 63.888

`--auto-overlap-frac <bool>` If set, GARLIC attempts to guess based on marker density.
Default: false

`--auto-winsize <bool>` If `--weighted` is set, guesses the best window size based on SNP density, otherwise initiates an ad hoc method for automatically selecting the # of SNPs in which to calculate LOD scores. Starts at the value specified by `--winsize` and increases by `<step size>` SNPs until finished.
Default: false

`--auto-winsize-coef <double1> ... <doubleN> <slope> <intercept>`: coefficients of $\text{winsize} = \text{slope} * \log(\text{density}) + \text{intercept}$, used by `--auto-winsize` with `--weighted`. The defaults (8.3235 138.0521) are an empirical fit to human SNP-array data; override them for other ascertainment schemes or species.
Default: 8.3235 138.0521

`--auto-winsize-step <int>` Step size for automatic window selection algorithm.
Default: 10

`--auto-winsize-threshold <double>` Smoothness criterion at which the `--auto-winsize` search stops. Lower values demand a smoother LOD score density and therefore a larger window.
Default: 0.5

`--autosomes-only <bool>` Drop sex chromosomes (X, Y, 23, 24) before calling. A hemizygous male genotype is written as a homozygous call in a TPED and is indistinguishable from true autozygosity, so a male X chromosome is called as one run spanning the whole chromosome. `garlic` warns when sex chromosomes are present; this removes them. Not the default, because it would change results for anyone already analysing them deliberately.
Default: false

`--build <string>` Choose which genome build to use for centromere locations: hg18, hg19, hg38 or t2t-chm13 (T2T-CHM13v2.0, UCSC hs1).
Default: none

`--centromere <string>` Provide custom centromere boundaries. Format `<chr> <start> <end>`.
Default: none

`--chr <string1> ... <stringN>` Analyse only these chromosomes, e.g. `--chr chr1 chr2 chrX`. Names are matched after normalisation, so '1' and 'chr1' are the same. It is an error to name a chromosome that is not in the data.
Default: all chromosomes in the input

--cm <bool> Measure ROH lengths in genetic distance units. This requires a mapfile.
Default: false

--dump-docs <string> Write the command line reference to stdout in a documentation format and exit: txt for the README block, tex for a LaTeX description list. Used by 'make docs' so the documentation cannot drift from the program.
Default: _none

--error <double> The assumed genotyping error rate.
Default: -1

--force <bool> Overwrite existing output files. Without this, garlic refuses to clobber an existing <out>.roh.bed.
Default: false

--freq-file <string> A file specifying allele frequencies for each population for all variants. File format: CHR SNP POS ALLELE FREQ <chr> <locus ID> <allele> <freq> By default, this is calculated automatically from the provided data.
Default: none

--freq-only <bool> If set, calculates a freq file from provided data and then exits. Uses minimal RAM.
Default: false

--froh <bool> Also write <out>.froh.tsv: per-individual autozygous total and fraction, broken down by ROH size class. The denominator is stated in the file header.
Default: false

--gl-type <string> Which genotype-quality field to use. With **--vcf** it is read from the FORMAT column; with **--tgls** it names the form of the single value per genotype in the .tgls file. The two sources are NOT equivalent. With **--vcf** (recommended): GQ error = $10^{-(GQ/10)}$, the VCF spec's definition directly. PL The whole PL array is read and converted to a posterior: $P(g)$ proportional to $10^{-(PL_g/10)}$, error = $1 - P(\text{called})/\text{sum}$. This is the only correct use of a VCF's PL. A VCF normalises PL so the CALLED genotype is exactly 0, so that one value carries no information – it is 0 for every call, confident or not. The information is in how much worse the alternatives are. GL The same, via $PL = -10 \cdot GL$. A missing field is an error naming whichever of GQ/PL/GL the file does carry, and a called genotype with '.' for the field is an error rather than a silent fall back to **--error**. With **--tgls** (legacy; the format holds ONE value per genotype): GQ As above. The only form a VCF's values can be dropped into. PL error = $1 - 10^{-(PL/10)}$, i.e. $P(\text{genotype CORRECT})$. *** NOT the VCF PL field. *** Under this definition PL=0 means an error of 0, and the integers map to 0, 0.21, 0.37, 0.50, ... – none a realistic error rate (0.001 would be PL=0.00435). Feeding VCF PLs here is rejected. Use **--vcf --gl-type PL** instead. GL error = $1 - 10^{GL}$, with the same caveat.
Default: none

--gmm-max-iter <int> Maximum EM iterations when fitting ROH size classes.
Default: 1000

--gmm-tol <double> Convergence tolerance for the ROH size-class EM fit.
Default: 1e-05

--help <bool> Prints this help dialog.
Default: false

--kde-cut <double> Extend the density grid this many bandwidths beyond the range of the data.
Default: 3

--kde-points <int> Number of grid points at which the LOD score density is evaluated. Also sets the resolution at which the between-mode minimum (the LOD cutoff) is located.
Default: 512

--kde-subsample <int> The number of individuals to randomly sample for LOD score KDE. If there are fewer individuals in the population all are used. The default, 0, uses every individual, which makes the selected LOD cutoff a deterministic function of the data alone. A positive value subsamples, which saves memory but makes the cutoff depend on **--seed**.
Default: 0

--kde-thin-step <int> Take every Nth LOD score window when building the KDE. 0 (the default) uses the window size, which keeps the sampled windows non-overlapping; 1 uses every window. Replaces **--no-kde-thinning**, which is equivalent to 1 and is kept as a deprecated alias.
Default: 0

--ld-subsample <int> The number of individuals to randomly sample for LD calculation during wLOD. If there are fewer individuals in the population all are used. Set ≤ 0 to use all individuals (will increase runtime).
Default: 0

--load-params <string> Read flag values from a <out>.params.json written by a previous run. Flags given on the command line take precedence over the file, so a run can be repeated with one parameter changed.
*Default: **--none***

--lod-cutoff <double> For LOD based ROH calling, specify a single LOD score cutoff above which ROH are called in all populations. By default, this is chosen automatically with KDE.
Default: -999999

--map <string> Provide a scaffold genetic map, sites that aren't present within this file are interpolated. Sites outside the bounds are filtered. This is required for wLOD calculations and any runs for which you wish to report ROH in units of cM.
Default: none

--max-gap <int> A LOD score window is not calculated if the gap (in bps) between two loci is greater than this value.
Default: 200000

--max-winsize <int> Upper bound on the window size that **--auto-winsize** will try. The search previously had no bound and would grow past the number of loci if the smoothness criterion was never met.
Default: 1000

--mode-smooth-span <int> Number of adjacent grid points used to smooth the density when locating its modes. Must be smaller than **--kde-points**.
Default: 20

--mu <double> Mutation rate per bp per generation for **--weighted** calculation.
Default: 1e-09

--nclust <int> Set number of clusters for GMM classification of ROH lengths.
Default: 3

--no-centromere <bool> Treat every chromosome as having no assembled gap, instead of requiring **--build** or **--centromere**. For organisms with no centromere gap in the assembly, and for references garlic has no built-in table for (only hg18, hg19 and hg38 are built in; T2T-CHM13 is not). Mutually exclusive with **--build** and **--centromere**.
Default: false

--no-kde-thinning <bool> Set this flag to send all LOD score data to KDE function. This may dramatically increase runtime.
Default: false

--out <string> The base name for all output files.
Default: outfile

--outdir <string> Write all output files into this directory, which is created if it does not exist. Equivalent to prefixing **--out** with the path.

--overlap-frac <double> The minimum fraction of overlapping windows above the LOD cutoff required to begin constructing a run. ROH will have a lower bound size threshold of $WIN_SIZE * OVERLAP_FRAC$. If set to 0, GARLIC sets the value to the lowest sensible value: $1/winsize$.
Default: 0.25

--phased <bool> Set if data are phased and you want to calculate r^2 instead of h^2 while **--weighted** is set. Uses extra RAM. Has no effect on computations without **--weighted**.
Default: false

--pop <string> A file mapping sample ID to population, replacing the population labels that would otherwise come from the TFAM. Whitespace separated, '#' comments and blank lines ignored, .gz accepted: <sample_id> <population> [sex] Two required columns in THAT order. Note this is the opposite order to a TFAM, whose first column is the population. The optional third column is sex in PLINK coding (1 male, 2 female, 0 or -9 unknown); unlike a TFAM, anything else is an error rather than being read as unknown. Rows are matched by sample ID, never by file order. Every sample in the data must have a row; extra rows are ignored and counted, so a cohort-wide file can be used against a subset.
Default: none

--quiet <bool> Suppress the progress bar and informational messages. Errors and warnings still go to stderr and to <out>.error.
Default: false

--raw-lod <bool> If set, LOD scores will be output to gzip compressed files.
Default: false

--resample <int> Number of resamples for estimating allele frequencies. When set to 0 (default), garlic will use allele frequencies as calculated from the data.
Default: 0

--seed <int> Random seed, for reproducible runs. Affects **--kde-subsample**, **--ld-subsample** and **--resample**. If 0 (default), a nondeterministic seed is drawn and written to the log file so the run can be repeated exactly.
Default: 0

--size-bounds <double1> ... <doubleN> Specify the size class boundaries ROH boundaries. By default, this is chosen automatically with a 3-component GMM. Must provide numbers in increasing order.
Default: -1.000000

--tfam <string> A tfam formatted file containing population and individual IDs.
Default: none

--tgls <string> A tgls file containing one per-genotype likelihood value per individual, in the same row order as the tped. See **--gl-type** for what the values must mean: for PL and GL this is NOT the VCF convention.
Default: none

--threads <int> The number of threads to use. Applies to LOD score calculation, the LD calculations under **--weighted**, the LOD score KDE, and ROH assembly.
Default: 1

--tped <string> A tped formatted file containing map and genotype information.
Default: none

--tped-missing <char> Single character missing data code for TPED files.
Default: 0

--vcf <string> A VCF file (plain or gzipped) to read genotypes from, instead of **--tped**. Biallelic SNVs with a GT field are used; multiallelic sites, indels and symbolic ALTs are skipped and the counts reported. Sample names come from the #CHROM line, so **--tfam** is neither needed nor accepted; use **--pop** to supply population labels. Sites must be grouped by chromosome. Ploidy must be 2 at every call, and with **--phased** every call must use '—'.
Default: none

--vcf-pass-only <bool> Skip sites whose FILTER column is neither PASS nor '.'. Without this, such sites are KEPT and their number is reported, because a FILTER column reflects the caller's thresholds rather than garlic's.
Default: false

--verbose <bool> Show the progress bar even when stderr is not a terminal. By default it is drawn only on a terminal, because it works by emitting backspaces.
Default: false

--version <bool> Print the version and exit.
Default: false

--weighted <bool> Compute LOD scores weighted by LD and probability of mutation.
Default: false

--winsize <int> The window size in # of SNPs in which to calculate LOD scores.
Default: 0

--winsize-multi <int1> ... <intN> Provide several window sizes (in # of SNPs) to calculate LOD scores. LOD score KDEs for each window size will be output for inspection.
Default: -1

-h <bool> Prints this help dialog.
Default: false

References

- Pemberton, T. J., Absher, D., Feldman, M. W., Myers, R. M., Rosenberg, N. A., and Li, J. Z. 2012. Genomic patterns of homozygosity in worldwide human populations. *American Journal of Human Genetics*, 91: 275–292.
- Purcell, S., Neale, B., K., T.-B., Thomas, L., Ferreira, M. A. R., Bender, D., Maller, J., Sklar, P., de Bakker, P. I. W., Daly, M. J., and Sham, P. C. 2007. PLINK: a toolset for whole-genome association and population-based linkage analysis. *American Journal of Human Genetics*, 81: 559–575.